

University of Portsmouth

School of Creative Technologies

Final Year Project

How approachable is scientific and mathematical academia to game developers that have little to no formal background

by

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Modern video game development increasingly relies on complex algorithms, physics simulations, and mathematical concepts. However, it could be argued that a significant gap between contemporary research and the skills of game developers who lack a formal academic background. This report investigates the accessibility of different scientific and mathematical literature from a game development perspective. To evaluate this, this study uses different approaches such as first hand research that was conducted through presenting developers with standard academic notation such as summation ($\sum$) and product ($\prod$) to assess their baseline comprehension. This report aims to highlight the need for more accessible academic resources to help bridge the gap between the academic and practical side of game development.

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Chapter 1

Introduction

While players primarily experience gameplay and immersive worlds, these experiences are often driven - behind the scenes - by complex algorithms and mathematical concepts.

As the creative and technical vision of the project expands, the developers reliance on advanced mathematics scales with it; whether it be to achieve more realistic physics, procedural generation, or optimise existing systems. However, when developers turn to contemporary research literature to implement or optimise these visions, they may face a critical roadblock. A significant gap exists between the formal academic notation that can be found in scientific papers, such as summation ($\sum$), or product ($\prod$) symbols, and the practical, programming-orientated skillset of developers who may lack an existing formal background in these fields of academia. Consequently, the literature that has been created to provide a way to establish and solve problems become arguably inaccessible to a large portion of the development community.

This report aims to measure a wide-range of developers' baseline comprehension of different common-place concepts and notation that can be found in mathematic literature, and comparing it to the level of explanation and context that can be found in these papers.

1.1 Defining the developer

In the context of this report, a 'developer' is someone who:

1. has written code in C++ *or in a comparable game-engine language such as C#*
2. has less than **2 years** of formal further/higher-education in mathematics
3. is current engaged in either **independant, hobbyist, or industry** game development *or currently engaged in similar fields such as software engineering or web development*

1.2 Defining the paper

In the context of this report, a ‘paper’ may refer to:

1. a report that is written specifically with game development in mind
2. a report that is written with physicists or mathematicians in mind
3. a report from adjacent technical fields, such as computer science, graphics programming or artificial intelligence, which uses any mathematical concepts.
4. any peer-reviewed academic literature, journal article, or any other formal academic writing that contains contemporary physics or mathematical content.

Chapter 2

Literature Review

Chapter 3

Methodology

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